

AUDIT-10-TERMUX_MAIN.md

Audit & Extraction Report: Subfolder 11 of _Lex-Novi-Æxentis-Copiæ (--• _TERMUX[>]_main.)

Scope: On-device Termux execution, DroidDesk cross-device workstation, post-reboot startup sequences, and GCP terminal authentication.

Parent Source: ___Lex-Novi-Æxentis-Copiæ/--•  _TERMUX_[>]_main.
(1XK7bRmY5yDJKDdUqwLtHw0dUmQ0zv-oq).

Status: All original files preserved 100% untouched.

1. Purpose & System Role: Mobile Shell & Workstation Engine

This folder contains the **concrete, battle-tested on-device deployment runbooks** for Node Alpha (Phone) and your Samsung Galaxy Tab S9 FE:

- Standardizes the automated startup sequence that launches local models and speech daemons on boot.
 - Establishes the **DroidDesk / Termux:X11 dual-device architecture** turning your phone into a headless compute server and your tablet into a desktop workstation.
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2. Major Technical Extractions & Discoveries

A. Post-Reboot Model & Voice Startup Sequence

Extracted directly from **Getting the local LLM + voice pipeline working (every time after a reboot)**:

1. Step 1 (Launch Local Model):

```
bash ~/aesop/deploy/phone/boot.sh
```

Spawns a detached **tmux** session named **llm** running Qwen 3.5 9B on the Hexagon NPU. Prints **AESOP Edge Online** in ~20 seconds.

2. Step 2 (Launch OpenWiki TUI):

ow

Launches OpenWiki pointed at the local NPU. Binds native hotkeys: **Ctrl+R** to record voice, **Ctrl+S** to transcribe via Moonshine STT and speak via Kokoro TTS (zero cloud latency).

3. **Step 3 (Claude Code Push-to-Talk):** Tap **ALT** then **K** (sending **Meta+K**) to trigger Claude Code's built-in voice input, bypassing Android keyboard input eating.
 - *Action:* Codified into **01_SOVEREIGN_NODE_AND_APK_TOPOLOGY.md** boot sequence.

B. The DroidDesk Tablet-to-Phone Workstation Pipeline

Extracted directly from **Termux:X11 desktop install** :

- **The Tablet (Main Native Screen):**
 - Runs **Termux:X11** for lag-free, hardware-accelerated desktop rendering.
 - Runs DroidDesk (**pkg install x11-repo && ./setup.sh**) providing a clean PC desktop interface with app launchers.
- **The Phone (Headless Server & Claude CLI):**
 - Runs Debian via PRoot (**proot-distro install debian**).
 - Runs Node.js LTS and **@anthropic-ai/claude-code**.
 - Broadcasts its desktop locally via **noVNC** / **websockify** on port 8080:

```
vncserver :1
```

```
websockify --web /usr/share/novnc 8080 localhost:5901
```

- **The Unified Workflow:** The tablet opens a browser window inside DroidDesk pointing to **http://<PHONE_IP>:8080/vnc.html**, monitoring phone CLI tasks while multitasking natively.
- *Action:* Enveloped into **01_SOVEREIGN_NODE_AND_APK_TOPOLOGY.md**.

C. GCP \$1,000 Vertex Credit Terminal Fast-Track

Extracted from **Google cloud API hooks guide through termux.:**

- Bypasses the complex Google Cloud IAM web console entirely.
- Enables APIs instantly via Cloud Shell:

```
gcloud services enable aiplatform.googleapis.com storage.googleapis.com  
documentai.googleapis.com
```

- Authenticates mobile Termux without JSON service keys using application default login:

gcloud auth application-default login

- *Action:* Ingested into `04_skills_runtime/extracted_tools/cli/gcs_sync.sh`.

D. Verified Hardware Configs (`ASSETS.txt`)

- Mined real configuration parameters:
 - `processor_config.json`: Gemma4AudioFeatureExtractor spectrogram parameters (128 features, 750 seq length).
 - `htp_backend_ext_config.json`: QNN HTP configuration verified against Hexagon v79.
 - `mcp.json`: Configuration for `agentmemory` MCP server.
 - *Action:* Codified into `02_wiki_md/architectures/snapdragon_npu.md`.
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3. What Was Combined & What Was Trimmed

- **Fluff Trimmed:** Trimmed debugging logs from broken third-party audio packages (`termux-tts-speak` hangs and broken symlinks).
- **Combined Runbooks:**
 - Unified the 2026-08-28 GLM Setup Guide with the post-reboot startup sequence into a master operator runbook in `02_wiki_md/runbooks/`.